

Lac Culture

Lac insects and their products have been known to naturalists since very early times. The lac has been referred in ancient Sanskrit works viz., Atharva-Veda (Dave, 1950; Hora, 1952) and was called as 'Luxa'. It is mentioned in Mahabharat that 'Luxa Griha' was made up of lac which was prepared by Kaurava for Pandavas. Abul Fazal (1590) in his famous book 'Ain-i-Akbari' has mentioned in detail about the lac industry in India. Mahdihassan (1950, 1952), has referred about the lac insect and its products in China. The first scientific reference regarding the lac and lac insect is the report of Kerr and Glover in 1782. Subsequently, much work has been done by various workers on the organization, distribution, taxonomy, host plants, culture, production, enemies, chemistry and technology.

Three products from lac insects, viz., the lac-dye, lac-wax and lac (resin) have been items of trade and commerce.

Distribution

India has its monopoly on the production of lac. Other countries like Africa, Australia, Brazil, Burma, Sri Lanka, China, Formosa, France, W. Germany, Japan, Malaya, Nepal, Spain, Thailand, Turkey, U.S.A. and some others also produce lac. But in Thailand, Malaya, Burma and Nepal the lac producing industries are increasing day-by-day. Thailand has become the main competitor of India in export of lac. In India major lac producing places are Assam (Kashi Hills), Bengal (Calcutta, Jangipur, Murshidabad, Mathrapur, Malda), Bihar (Manbhum, Palamau, Ranchi, Santhal Pragana), Delhi, Gujarat,

Hyderabad, Kashmir, Madhya Pradesh (Damoh, Champa, Bilaspur, Rewa, Umaria), Chennai, Coimbatore, Mysore, Orissa (Cuttak, Mayurbhanj), Punjab (Hoshiarpur, Shahpur), Rajasthan (Indergarh, Kota, Jaipur, Jhallawar, Karauli), and Uttar Pradesh (Ghazipur, Mirzapur, Agra) etc.

Lac Insect (Lakh ka-kira)

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Sub-order	Homoptera
Super-family	Coccidae
Family	Leciferidae
Genus	<u>Tachardia</u>
Species	<u>lacca</u>

Arthropod
Insecta

Lac insect (*Tachardia lacca*) previously known as *Laccifer lacca* is a minute, resinous, crawling scale-insect which inserts its beak into plant tissues, sucks juices and grows, and secretes lac from the hind end of the body. Its own body ultimately gets covered with lac in the 'CELL'. Lac is actually secreted for its protection and not for the food of the insect. The commercial lac is produced in large quantities by female as a protective covering of its body which is injurious to the host plants.

Male. Male is red in colour and 1.2 to 1.5 mm in length. It secretes bright creamy lac. It has reduced eyes and ten segmented antennae. The mouth-parts are of piercing and sucking type. Thorax bears three pairs of legs and one pair of hyaline wings. The abdomen is eight segmented and terminates into a short, chitinous prominent general sheath containing penis. On either side of this genital sheath a white elongated caudal seta is found.

Y/p
30 p

Female. Female is larger than males and measures about 4 to 5 mm in length. The pyriform body of the female is enclosed in a resinous cell. The head, thorax and abdomen are not clearly distinct. The mouth-parts are of piercing and sucking type. The antennae are clearly visible and degenerated. The posterior end of the body has a median and two lateral processes. The legs are in degenerated form.

Life History

Each mature female just after fertilization lays about 200 to 500 eggs in a cell in which she is enclosed. The oviposition takes place into the incubating chamber which is formed by the contraction of the body of the female in forward direction inside the lac cell. The eggs are laid in the months of October and November. After six weeks of laying, the eggs are hatched into first instar nymphs in the months of November and December. When nymphs emerge they are in quite large number. This mass emergence of the nymphs is known as 'SWARMING' (Fig. 1).

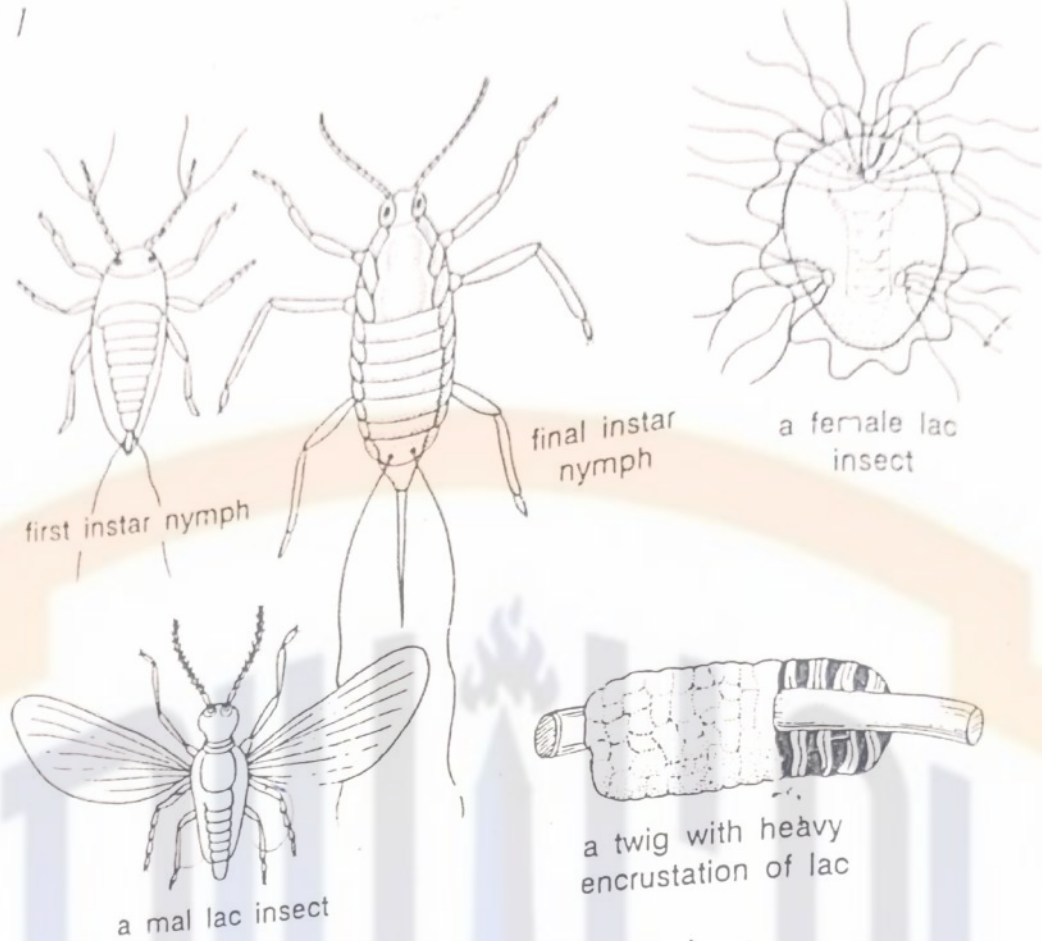


Fig. 1. Life history of *Tachardia lacca*.

Nymph. At the time of emergence the nymphs are about 0.5 mm in length, red coloured and boat-shaped. The head bears paired antennae, ocelli and ventrally situated piercing and sucking type of mouth-parts. The mouth-parts are provided with proboscis. The three segmented well developed thorax contains two pairs of spiracles and only one pair of walking legs. The abdomen contains two pairs of legs and terminates into a pair of long caudal setae. The active nymphs can crawl to a considerable distance so, just after emergence they start moving in search of food and reach their host plants, preferably on young and succulent shoots because the young nymphs are unable to settle and feed on hard twigs. These nymphs settle very close to each other on the twig of the host plant which further collapses completely and forms a continuous covering even on the lower surface of the twig. The number of nymphs that settle per square inch area is about 150 to 200. Settled nymphs suck the sap from the twig of the host plant and start to secrete the resinous substance by special dermal glands which are located all over the body. As the resinous secretion comes in contact with air, it soon becomes hard and forms a coating over the body of nymph and is called as 'CELL'. Within this cell various life processes like growth of the nymph, morphological changes and lac secretion take place.

The male 'Cell' is elongated and cigar-shaped having two holes i.e., anterior and posterior. From the posterior hole which is covered by a flap or operculum.

The male insect comes out by pushing open the operculum. After six to eight weeks of stationary life the nymphs are metamorphosed as a result of which some (30%) active winged males and maximum (70%) emerge in the form of females which are wingless. The females get fixed on the host plant in resinous mass. The males walk over the encrustations of females and fertilize them within their oval cells through anal opening. The males leave the parent cell after fertilizing the female. One male is capable to fertilize many females. The female nymph once settled never moves but undergoes 3 moults inside her cell losing its eyes and legs, and with rudimentary antennae only. The fertilization of female is followed by a rapid growth of the female body till it begins to lay eggs in October and November. From these eggs male and female emerge in February to March. The male fertilizes the females of this generation and the fertilized female lays eggs in months of June to July and dies secreting lac all the time. Thus, the life cycle reoccurs twice in one year on the same host plant.

Due to short life period males do not take major part in the secretion of lac but female secretes lac throughout her life and its life span is longer than males. Major quantity of lac is secreted from females. The life cycle period depends mainly on ecological factors of the region.

Host Plants

The lac insects have more than one host plant. The selection of suitable host plant for the cultivation of lac is of much importance. To establish the lac industry one should know well about the topographic and climatic conditions for the growth of host plants suitable for that particular region. Brun (1958) has mentioned that 113 varieties of host plants are found in the geographical Indian regions including Pakistan and Burma. Out of these 113 host plants only 14 are very common in India which are as follows :

1. Kusum ✓	—	<i>Schleichera oleosa</i>
2. Babul ✓	—	<i>Acacia nilotica</i>
3. Ber ✓	—	<i>Zizyphus mauritanias</i>
4. Palas	—	<i>Butea monosperma</i>
5. Ghont	—	<i>Zizyphus xylopyra</i>
6. Khair	—	<i>Acacia catechu</i>
7. Peepal	—	<i>Ficus religiosa</i>
8. Gular	—	<i>F. glomerata</i>
9. Pakapi	—	<i>F. virens</i>
10. Putkal	—	<i>F. globella</i>
11. Mango ✓	—	<i>Mangifera indica</i>
12. Sal	—	<i>Shorea robusta</i>
13. Shisham ✓	—	<i>Dalbergia sisso</i>
14. Fig ✓	—	<i>Ficus carica</i>

The quality of lac is directly related with the quality of host plant. So far, no artificial product has been able to replace the lac. Khair, Kusum and Babul give better quality of lac when sown directly in the field. But Palas, Ber and

Ghont give good crop when they are first sown in nursery and then transplanted to the lac growing field. Palas and Ber produce a particular type of lac which is called as 'KUSUMI LAC'.

Cultivation of Lac

Lac cultivation is a complicated process, so the cultivators should know well about the inoculation, swarming period and harvesting of lac.

Inoculation

The first procedure in the lac cultivation is the inoculation of lac insect. Inoculation is the process by which young ones get associated properly with the host plant. Inoculation is of two types —

1. **Natural inoculation.** The inoculation taking place in normal routine or in natural way is very simple and common process during which the swarmed nymphs infect the same host plant again and start to suck the juices from the twigs. The natural incubation of swarmed nymphs has some drawbacks which are as follows —

(a) *Incomplete nutrition.* Lac insects with their piercing and sucking mouth-parts, pierce into succulent twigs and suck the cell sap of the same host plant for nutrition. If the cell sap of the same host plant is further sucked out by the swarmed nymphs of the second crop continuously, the growth of the host plant would be retarded. In this way lac insect may not be able to get enough nutrients from the same host plant. The lac insects due to lack of sufficient nutrients lose their proper development, thereby affecting the production of lac also.

(b) *Irregular inoculation.* During the natural inoculation it is not sure that uniform sequence of inoculation takes place. If inoculation is not of continuous fashion, a regular crop of lac may not be obtained.

(c) *Unfavourable climatic conditions.* At the time of swarming a number of factors like high intensity of sunlight, heavy rainfall, flow of wind etc. affect the proper inoculation of nymphs. These natural environmental factors may also affect the host plant at the same time and may cause a gap of inoculation resulting in irregularity of the lac crop.

(d) *Multiplication of parasites and predators.* Lac insects have certain enemies in the form of parasites and predators. If the crop is not harvested in time and lac is allowed to remain on the same twig, the multiplication of parasites and predators takes place which hampers the population growth of lac insects.

Thus, keeping in view the above drawbacks the natural procedure of inoculation is avoided and certain devices have been developed to ensure artificial method of inoculation.

2. **Artificial inoculation.** The main idea behind the artificial method of inoculation is to check all possible drawbacks of natural inoculation.

In this method first of all host plant should be pruned in January or June. The twigs bearing insect nymphs which are about to swarm, or just before swarming are cut in sizes ranging between 20 to 30 cm in length. Then the cut pieces of these twigs are tied to fresh trees in such a way that each stick touches the tender branch of the tree at several places which form bridges for the migration of the nymphs. After swarming, these twigs should be removed and separated from the host plant. The following precautions should be taken in artificial inoculation :

(i) One must ensure that the twigs, which are going to be tied on fresh host plant, are having good number of nymphs or eggs. It is also possible that from many of the twigs nymphs have swarmed out, thus inoculation would prove unsuccessful. (ii) The twigs provided with eggs or nymphs should be without any parasite and predator. (iii) The eggs or nymphs present on the twigs should be healthy and about to swarm so that one has not to wait for longer period and thus save time. (iv) For the uniformity of inoculation, 3 to 4 twigs should be utilised. (v) Host plants should be changed from time to time for the proper nutrition of the nymphs.

→ These insects are very small and if they move to a long distance there are chances of mortality of the nymphs. Due to maximum contact of twigs, swarming nymphs have not to move for long distance and find suitable places to establish on the host plant.

Inoculation period

In India two types of crops viz., Rangini and Kusumi are grown in a year. The Rangini crop is of two types called as Kartiki and Baisakhi crop which produce Kartiki and Baisakhi lac respectively. The Kusumi crop is also of two types viz., Agahani and Jethi which produce Agahani and Jethi lac respectively.

Thus, the inoculation periods of all the four types of crops are different. The inoculations of Kartiki, Baisakhi, Agahani and Jethi crops are recommended in months of June to July, October to November, July and January to February respectively. But if continuously four crops are taken, the plant would not get any rest which may cause less production of lac.

Swarming

It is very important phase in the life history of lac insect. So one should have accurate knowledge about the actual date of the swarming. At the time of swarming, the upper surface has yellow spot on the anal region. At this stage muscle contracts and insect gets detached from the place of attachment. Thus, it leaves a hollow cavity which later on gets covered with wax also. When these eggs are to be hatched out they become orange coloured. Thus, it is an indication that swarming has taken place. Thus, by trials and learning

methods i.e., by practice one could know about the exact date of swarming by looking at the colour of the eggs.

Harvesting of lac

The process of collection of ready lac from host tree is known as harvesting. In common practice the harvesting is of two types.

1. **Immature harvesting.** The harvesting of the lac before swarming is called as immature type of harvesting and the lac thus obtained is known as **'ARI LAC'**.*

2. **Mature harvesting.** The collection of crop after the swarming is called as mature harvesting and the lac obtained is known as **'MATURE LAC'**.

The harvesting of lac before the swarming has some drawbacks because the lac insects may be damaged at the time of harvesting which would affect the population of lac insects and ultimately result in great economic loss to the cultivators. But in case of palas lac (Rangini lac) it is found that Ari lac gives better production. Therefore, Ari lac harvesting is recommended in case of palas only. In all other cases immature harvesting should be discouraged. It is also found that in cold areas mature crop yields better quality of lac.

Harvesting period. The harvesting periods of different crops are quite different in accordance with the inoculation of crops. Kartiki crop is harvested in October to November whereas, Baisakhi crop in May and June. The other crops like Agahani and Jethi are harvested in January to February and June to July respectively.

Recent Plan for Lac Cultivation

With the increasing number of lac industries some advanced plans have been recommended for the better cultivation of lac crops. Two types of planning are used now-a-days.

1. **Ceupe system.** All the trees of host plants of a definite area are not used under continuous cultivations process of lac crop because if all host plants of a farm would be under continuous attack of lac insects, 100% plants may not get any rest and thus the production of the lac would be affected due to deficiency of nutritive cell sap to the swarmed nymphs and adults. So, the plants of a farm are numbered into 5 groups of plants. This artificial division or marking of trees is called as ceupe system of crop cultivation. In this system when one group of host plants is under the process of cultivation of lac, other groups of host plants would be under rest.

2. **Alternation of plant.** In this system the variety of host plant is changed after one crop. So, swarmed nymphs are inoculated on the tree of other variety of host plant. In this way every host plant can get enough rest resulting into better production of lac.

(Z-18)

Processing of the Lac Industry

When the crop matures fully most of the lac is harvested and some part is left on the host plant. For the proper cultivation, the host plant should be pruned in January every year.

→ The twig bearing the lac along with eggs is called a BROOD LAC STICK and lac is known as BROOD or STICK LAC. The processing starts with the scraping of the stick lac from the twig. The scraped lac is subjected to removal of many impurities like dead parts of the lac insects, eggs and colouring matter, and finally crushed by hand-operated mortars. Then the material is air dried and obtained in the form of granules which is known as SEED LAC. This seed lac is soaked in water, washed, dried in sun light, bleached and heated to melt on charcoal fire in cloth bag of 3 to 4 metre. At the time of heating the bag is twisted and the lac is squeezed out of the bag. The impurities of the lac are left out in the bag, and are called as KIRRI LAC. The squeezed lac is now allowed to cool and solidify around the button-shaped forms which is now called BUTTON LAC or PURE LAC. This pure lac when stretched into thin sheet is called as SHEET LAC. This sheet lac when dissolved in water, produces white or orange coloured lac which is called as SHELL LAC. Shell lac is, in fact, prepared by boiling the seed lac with yellow arsenic in a certain proportion. Thus the shell lac is most purified form of lac.

The quality of lac depends upon the host plant. Kusumi lac is said to be the best lac while Dhak is supposed to be the worst and cheapest one. The quality and colour of the lac is variable according to the presence of gum and resins in the host plants.

Composition of lac

Lac is a complex substance having large amount of resins, together with sugar, water and other alkaline substances. The percentage of various constituents are as given below :

(1) Resin	—	68 to 90%
(2) Dye	—	2 to 10%
(3) Wax	—	6%
(4) Albuminous matter	—	5 to 10%
(5) Mineral matter	—	3 to 7% and
(6) Water	—	3%

Properties of lac

- (1) Lac is not soluble in water but easily soluble in alcohol. This property of lac has great value for insulation of electrical connections.
- (2) Lac is easily fusible on heating.
- (3) Lac has adhesive quality.

(Z-18)

- (4) It has binding property when mixed with alcohol.
- (5) Lac is also soluble in weak alkali like ammonia.
- (6) Lac is a bad conductor of heat.

Enemies of Lac Cultivation

Lac cultivation is destroyed by biotic and abiotic factors :

1. Abiotic enemies . These are high intensity of light, high temperature, high humidity, heavy rainfall and flow of wind.

2. Biotic enemies. The main biotic enemies of lac cultivation are mammals and insects. Krishnaswami et. al. (1957, 59), and Gepulpure et. al. (1963), have reported that squirrel, rats, and monkeys cause great damage to the lac crop.

The insects are very powerful enemies of lac crop. Annual loss due to the insect enemies is to the tune of about four lakh maunds. The insects damage the crops in different ways.

1. Parasites. The lac insects are parasitised by eight species of chalcidoid parasites like, *Parenchthrodryinus clavicornis*, *Erencyrtus dewitzii*, *Tachardiaephagus tachardiae*, *Tachardiaephagus tachardiae* var. *somervilli*, *Eupelmus tachardiae*, *Coccophagustschirchii*, *Mariettajavensis* and *Tetrastichus purpureus*. These parasites lay their eggs into lac insects and parasitised 4.8 to 9.9% of lac insects per year and 1/3 of the parasitised cells are males. Thus, it may be concluded that parasitization is not a major cause of the damage to the lac cultivation.

2. Predators. Predators cause very severe damage to lac cultivation and 35% of the lac cells are damaged by two predators viz., *Eublemma amabilis* Moore (Lepidoptera : Noctuidae) and *Holocera pulverea* (Meyr) (Lepidoptera : Blastobasidae). Female lays eggs near encrusted twigs from where larva emerges and feeds on lac insects.

Precautions

- (1) Twigs for inoculation should be cut just before the swarming to get healthy brood.
- (2) Twigs used for inoculation should be free from predators and parasites.
- (3) Twigs tied for inoculation should be removed from inoculated host plants after a maximum period of 20 days.
- (4) Lac left on the host tree for swarming should be removed in October and November.
- (5) The brood lac after swarming should be destroyed along with predators and parasites on it.
- (6) The lac scraped from the tree should be taken away from the area of lac infected trees.
- (7) Fumigation and water immersion of lac, before removing from twig, should be done.

Lac Industry in India

India used to produce about 97 per cent of the total lac output in the world but at present it has come down to 50-60 per cent. The cultivation of lac has been good source as an earner of foreign currency. About 50 per cent of the total lac produced in India is obtained from Chhotanagpur area. States like Orissa, Punjab, Madhya Pradesh, W. Bengal, Uttar Pradesh, Gujarat, Rajasthan, Assam etc. are increasing the production of lac now-a-days. On a very small level lac producing is also reported from Delhi and Kashmir. The average yearly yield of lac in India is about 15,000 metric tons. A lac research institute 'Indian Lac Research Institute' Namkum, Ranchi had been established in 1925 which is producing good quality of white lac. The Indian white lac is supposed to be better than red or other coloured lac because they produce stain or spots at places where they are kept. This is mostly small scale industry with around 350 factories, mostly located in Bihar. In Mirzapur district alone there are about 40 factories. Out of total lac produced in India about 85 to 95 per cent is exported specially to Britain, U.S.A., Russia and West Germany.

Economic Importance

In 19th century lac dye was in more use than lac resin. Presently due to availability of a better and cheaper annaline dyes the use of lac as a dye has been discarded. The manifest uses of lac is one of the Nature's standing gifts. The various used to which it is put are —

- (1) It is utilized in the preparation of gramophone records. Previously this industry utilized major part of the lac produced annually. But now a days to a great extent plastic is being used in this trade.
- (2) It is of utility to Jewellers and Goldsmiths who use lac a filling material in the hollows in gold ornaments like bracelets, armlets and necklaces etc.
- (3) It is an essential gradient used extensively for making polishes, paints and varnishes for finishing wooden as well as metal furnitures and doors etc.
- (4) It is utilized for the preparation of toys, buttons, in pottery and artificial leather.
- (5) It is used in the manufacture of photographic material, lithographic ink and for stiffening felt and hat materials.
- (6) It is used as an insulating material for electrical goods. *Substance at boat hull to prevent fouling*
- (7) It is also used in confectionary trade as antifouling for applying on ship bottom, grinding stone industry and for ammunition and fire works.
- (8) Last but not the least used commonly as sealing wax. *Sealing wax*

Thus, it is of great use and considered to be as one of the cash crops for the cultivators and also to the Government as source of foreign exchange earners which amount to crores of rupees annually.

Pearl Culture

A large number of molluscs are found on the land and in the water reservoirs like sea, fresh water and brackish water. They are used for food, for shells and in pearl industries. Pearl is a white, highly shining globular concretion found within the shell of an Oyster (a kind of shell-fish). It is also called as 'Moti' or Muktafal'. Since antiquity, pearl has been reputed as one of the rarest gems and was previously imagined as the tears of moon but the reality, is far away than imaginary flight of poets. In Sanskrit literature it is mentioned that during 'Swati Nakshtra' when a drop of water falls in between the mantle and the nacre layer, pearl develops which seems to be more reasonable. The rare occurrence of pearl in the nature is due to its peculiar mode of formation which may not happen frequently in the natural conditions. Pearl is known to Chinese since 2300 B.C. Kautilya and Shakespeare in their literature have mentioned about the pearl in detail. The researchers engaged in pearl culture organisation have planned to stimulate this natural phenomenon and have succeeded in producing cultured pearls exactly of the same substance as the natural pearl.

History of Industry

For the first time the idea of pearl industry was evoked in Japan which was carried out in the Bay of Japan located at South coast of Hansoo. But in Japan, pearl culturists feel difficulty due to unfavourable climatic conditions. Kokichi Mikimoto (1858-1954) is referred to be the father of pearl industry in Japan. In 1890 an exhibition on Annual Domestic Industrial Promotion

was organised in Tokyo where among a number of industrial products pearls were sold at very high prices. Mikimoto came to know that pearls can only be formed in nature and not by artificial devices. This exhibition encouraged him to manipulate the oysters to do what they did in nature. In 1890 he established a pearl farm on small Island of Taba and began to culture oysters but could not get success for two years. In the third year due to a red-tide his farm was destroyed completely except for a batch of some oysters which survived. Subsequently on 11th July 1893, his wife Uma when opened an oyster shell to her surprise and joy she found a pearl duly formed in it. It was a memorable day for them. Then in 1896 he got a patent for pearl culture. Tokichi Nishikawa, a Govt. scientist of Misaki Marine Biological Laboratory of Tokyo University, was the first person to get spherical artificial pearl.

Pearl Producing Molluscs

Although a number of bivalves have ability to produce pearl under suitable climatic conditions but high quality of pearls are obtained from pearl oysters of Genus *Pinctada roding* belonging to class—Bivalvia, family—Pteriidae. A number of species of this Genus like, *P. vulgaris* (Schumacher), *P. chemnitzii* (Philippi), *P. margaritifera* (Linn.), *P. anomioides* (Reeve) and *P. atropurpurea* (Dunker), are found in Indian water resources. *P. vulgaris* is a common oyster distributed in the gulf of Kutch, gulf of Mannar and the Pak bay.

Apart from the true pearl oyster belonging to the genus *Pinctada* a large number of other marine and a few fresh water molluscs are also found to produce pearls or pearl like concretions. These are Ear-shell (*Haliotis* Linn). Sea muscle (*Mytilus* Linn) and windowpane oyster (*Placuna blacenta* Linn). *P. margaritifera* and *P. maxima* are giant species and produce pearls of bigger size but of inferior quality.

Pearl Producing Sites in India

Mostly the pearl oysters are inhabiting on the ridges of rocks or dead corals (secreted by many species of polyp), forming extensive pearl banks or at the depth of 18 to 22 metres at a distance of 19 km from the shore. The pearl oyster beds of the east coast are more extensive and productive than those of the west coast. These pearl beds produce best quality of pearls called as 'Lingha Pearl'. The pearl oysters are obtained from the reefs in the gulf of Mannar, gulf of Kutch, Pak bay and Baroda.

Pearl Formation

Pearl formation is an interesting phenomenon for protection against foreign invaders, parasites, sand grain, small broken twigs of sea-seeds or a small insect accidentally entering the body of the oyster which happens to adhere to a part of its mantle (Fig. 1). The mantle epithelium at once encloses it like



Fig. 1. Diagrammatic sketch showing site of pearl formation in *Mytilus*.

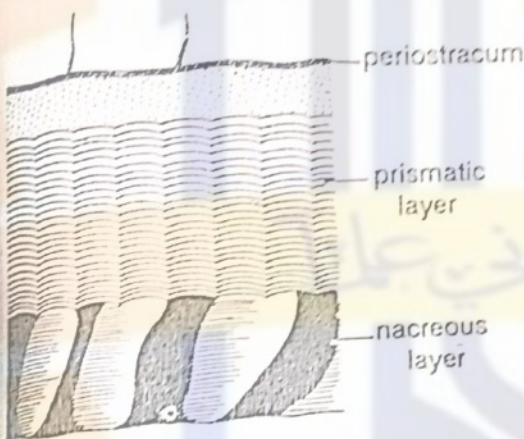


Fig. 3. Vertical section through the shell.



Fig. 2. Stages in pearl formation : A. Primary stage ; B. progressive ; C. Final stage.

a sac and starts to secrete concentric layers of nacre around it from defence point of view and completely encloses it gradually. Nacre is secreted continuously by the epithelial layer of the mantle and deposited around the foreign particle in the form of several layers and ultimately the pearl is formed (Fig. 2).

For a detailed information, vertical section of the shell if cut, shows that it consists of three layers (Fig. 3).

1. Periostracum. It is an outer layer formed of a horny conchyolin which is a substance related to chitin. On its inner side the next layer is found which is known as prismatic layer.

2. Prismatic layer. It consists of a small prism-like deposit of calcite (Burnt ash) separated by thin layer of conchyolin. This layer is followed by the third layer known as nacreous layer.

3. Nacreous layer. It is the inner-most layer formed of calcium carbonate and called as 'Mother of the pearl'. It consists of alternating layers of calcium

carbonate and conchyolin arranged parallel to the surface. This nacreous layer is secreted by the entire outer surface of the mantle, while the first two layers are secreted only by the edge of the mantle.

Programming of Pearl Industry and Artificial Insertion of Nucleus

Although pearl industry may be established only on natural basis of pearl formed by oysters in the natural conditions but an artificial device to insert the nucleus as foreign particle in the shell of oyster has proved useful for the production of pearls in greater number. This whole process is very much complicated, technical and time taking and can be managed as given below—

[I] Collection of oysters

The oysters are collected from the bottom sea by the divers particularly women divers in Japan who are called as 'AMA' which in Japanese means 'the girls of Sea.' The divers usually have got training for proper diving into the sea water since their childhood for the search of sea shells or sea weeds. The well protected suits of cotton with cap are used at the time of diving. Each diver has a small hand net at the time of diving when she goes upto 5 metre depth. The net helps in the collection of oysters from the bottom. The oysters collected by nets are stocked in the wooden bucket attached to the diver's left wrist by a cord and the diver with bucket comes up on the surface of water. An experienced diver can remain under water up to about one and half minute and can collect 2 to 10 oysters per dive. The best time for diving is from the early morning to mid day. The best period for the collection of the oysters is of two months in the summer season when the water is nearer and the sea is calm.

→ During diving in deep sea water the divers operate directly from the side of boat and a rope remains tightly fitted to the diver's wrist through which the operating boatman pulls the diver out with force up to the surface after receiving any signal from the partner. Thus, the whole collected oysters are stored and stored out. The oysters of same age group are segregated and two years old are kept in shallow water for future. Three years old oysters are sent to shallow water and in the months of April and May they are taken out. For pearl industry and proper supply of oyster, its eggs are incubated artificially which solves the problem of obtaining oysters for pearl culture.

Oysters are also caught by special type of cages ($84 \times 54 \times 20$ cm) by covering a heavy wire frame with two centimetre wire mesh. This cage is dipped into hot coltar as a measure against corrosion. Now this cage is dipped into the sand-cement mixture providing rough surface to the cages to which free swimming sopts get easily struck up. These cages are suspended at a depth of

metre from July to November where spots are easily available. These collected oysters are now transferred to rearing cages.

II] Rearing of oysters

The collected oysters are stocked and reared in special type of cages called s rearing cage. These cages are almost similar to those of collection cages except that they are further divided into 4 to 6 smaller chambers and lack the diagonal sub-divisions. They are also covered with metal mesh and with netting of cotton. These cages are well protected from natural enemies of oysters like Octopus, Eel, Devil fishes etc. The collected oysters are first cleaned and then placed into the culture cages for a period of about 10 to 10 days to recover the strain due to excessive handling and for the physiological adjustment to the shallow water conditions.

III] Insertion of nucleus

The insertion of nucleus as foreign particle is very much technical process and is of great importance for pearl industry. A number of methods are devised but most practicable and efficient method is one adopted by Nishikows. In this method a piece of mantle of living oyster is cut off and inserted together with a suitable nucleus inside the living tissue of another oyster. Following steps are taken for the insertion of nucleus.

1. Fitness of oyster for operation. The selected oysters for the insertion of nucleus should be healthy and strong enough to over come the shocks during operation. It is suggested that if the ovary and testis of oysters are got rid off they would be more resistant to the shocks of operation. For this purpose oysters are dipped into cold and warm current of water alternately which initiates them to eject their sperms and eggs in case of males and females respectively. Before operation, oysters are kept under stress of suffocation as a result they start to open their shells and at once a bamboo peg (piece) is inserted between the gap of two shells due to which shells may not be closed again.

2. Preparation of graft tissue. The piece of tissue which is inserted inside the mantle is called as 'GRAFT' tissue. A strip of about 7×0.75 cm is cut from the edge of mantle of healthy oysters by sharp knife. This piece is smoothed, cleaned and washed off the adhering mucus and again wiped off by wet sponge. The border of gill piece is removed by sharp scalpel. Now this tissue is trimmed to 2 to 3 cms long narrow strip and again cut transversely into small squares according to the size of the nucleus for insertion. These squares are kept in sea water at 22°C where they can survive for about 48 hours. The outer edges of these graft squares must be known because nacre secreting cells are found only on the outer surface of the mantle so it is essential to keep the outer surface in contact with the inserted nucleus.

3. Preparation of nucleus. Although any small particle may function as nucleus to initiate the pearl formation but it is reported that calcareous nucleus is the best because the deposition of nacre was found to be more satisfactory on calcareous nucleus as compared to any other particle. Best nucleus is formed by the shell of molluscs with heavy deposition of calcareous shells. Such type of molluscs are easily available in India but Japan depends on U.S.A. for good quality of calcareous shells. It is also notable that spherical nucleus is best for the formation of good quality of spherical pearl.

4. Insertion of nucleus. For the insertion of nucleus, oysters are fixed in a desk clamp in the position of right valve facing upward. Mantle folds are smoothly touched to expose the foot and the main body mass, followed by an incision into the epithelium of the foot and a slender channel into the main mass. Suddenly one graft tissue piece is placed into the channel and the nucleus is placed over the graft tissue which functions as a bed for the nucleus. Now the bamboo peg is quickly removed and oyster shells are closed automatically. For the insertion of the second nucleus similar operation is performed from the left side in the gonadial tissue and third insertion should never be tried. In Japan one trained girl can operate 25 to 40 oysters per hour and these girls are called as 'TOMARINE SON' means 'Miss Nucleus Pusher'. The operation period should not increase beyond 30 minutes and the oysters can not survive beyond one hour of the operation period. So operation and insertion of nucleus should be performed by experienced persons.

5. Post operational care. Nucleated oysters are placed into cages and suspended into sea water and attached with floating rafts to a depth of 2 to 3 metres for about 6 to 7 days to recover from the shocks due to operation. This period of 6 to 7 days is known as 'RECOVERY PERIOD.' Now oysters are examined properly and dead individuals are removed from cages. Sometimes, few oysters expel out the nucleus from the body due to heavy shock. Now-a-days it is examined by X-rays whether oysters are having inverted nucleus or not.
 -> About 3000 to 3600 nucleated oysters are kept in different cages suspended in sea water at 2 to 3 meters depth for 3 to 6 years and undisturbed except at the time of clearing and inspection. The pearl oysters grow best in warm shallow waters generally not more than 14 metre deep.

[IV] Harvesting of pearl

Pearls are harvested in the month of December to February which may slightly vary according to the climatic conditions of the industrial area. After the completion of 3 years of the insertion of nucleus, pearl oysters are harvested from the sea and the pearls are taken out from the shell.

[V] Clearing of pearls

After taking out the pearls from the oysters shell they are washed properly, cleared with the soap solution, but pearls should not be rubbed much.

Composition of Pearl

Pearl comprises of water, organic matter, calcium carbonate and the residue.

(1) Water : ✓	2-4%	(2) Organic matter : ✓	3.5-5.9%
(3) Calcium carbonate : ✓	90%	(4) Residue : ✓	0.1-0.8%

Quality of pearl

The pearls obtained are of variable shapes and sizes. They may be white, or cream red or pink red in colour. The spherical pearls of rainbow colour are rarely found. The best quality of pearl is known as 'LINGHA PEARL' and obtained from marine oysters. Pearl obtained from freshwater bivalves are not as valuable as those obtained from the marine oysters (Mishra, 1961).

Problems of Pearl Industry

Theoretically pearl culture appears to be very much easy but practically several problems crop up during culture. Number of enemies like eel, octopus, devil fishes etc, destroy the oyster. The lethal cold water, low salinity of water, turbidity of water and high range of temperature variation hamper the pearl industry by affecting the proper rearing of the oysters.

Suggestions for Pearl Industry ✓

- (1) During pearl fishing the nets used should be of such size from which smaller than a limited sized pearl may not be caught but should pass through the net to get a chance to grow.
- (2) The fishing of pearl oysters should not be much so as to exhaust the stock soon.
- (3) During the (breeding season) oysters should not be disturbed by fishing, rather allowed to breed freely. This period may be declared as close season for any fishing.